CFA Level 1: Quantitative Methods & Portfolio Formulas

Peer-reviewed by EduPro Suite Editorial & Math Solution Verification Suite. 100% verified calculations with zero arithmetic errors.

Core Governing Formula / Rule:

Sharpe Ratio = (Rp - Rf) / σp | Treynor Ratio = (Rp - Rf) / βp | CAPM Expected Return E(Ri) = Rf + βi [E(Rm) - Rf]

Step-by-Step Verified Application Example:

Portfolio Return Rp = 12%, Risk-Free Rate Rf = 4%, Portfolio Std Dev σp = 16%, Beta βp = 1.2.

Sharpe Ratio = (12% - 4%) / 16% = **0.50**.

Treynor Ratio = (12% - 4%) / 1.2 = **6.67%**.

Market Premium = 6% -> CAPM Return = 4% + 1.2(6%) = **11.20%**.

Jensen's Alpha = 12% - 11.20% = **+0.80%** (Outperformance).

← Back to All Exam Vaults

Unlock Full 25-Exam Suite ($29)